AINS HACKATHON

Technical Specification

AI for Entrepreneurship

Unlocking the entrepreneurship ecosystem through AI

Organised by **PNUD, GEWEET, ODC, IEEE SECTION, APII** and **AINS 4.0**

Contents

1 Global Requirements

1.1 Context & Mission

The central mission is to build AI systems that unlock the entrepreneurship ecosystem – removing structural barriers across orientation, diagnosis, financing access, and business evaluation. All use cases in this brief share the same foundational mission: *move beyond generic AI assistants and build systems that produce concrete, actionable outputs for real users.* Whatever the technical approach, the output must demonstrably solve a real problem for an entrepreneur in Tunisia.

> **Primary context:** Entrepreneurship in Tunisia and the MENA region. Every solution, regardless of use case, must be designed with this population as its primary user.

1.2 Technical Freedom

There is no imposed technology stack. Teams are free to choose any combination of tools, frameworks, models, and databases that best serve their use case. The quality of the solution and its real-world impact matter – not which specific library or platform was used.

Category	Details
AI / ML	Any approach: LLMs, classical ML, embeddings, rule-based engines, hybrid systems, or any combination
Data Stores	Any database: SQL, NoSQL, vector DB, graph DB, document store, or a combination
Frameworks	Any programming language, ML framework, cloud service, or on-premise deployment
Language (UI)	French and/or Arabic strongly preferred for Tunisian user-facing components
Data	Public data, synthetic data, and realistic mock data are all acceptable and must be documented (Preferably provided by PNUD)

1.3 What a Good Solution Must Achieve

1. **Real-world relevance.** The system addresses a concrete, documented pain point for entrepreneurs in Tunisia – not a hypothetical scenario.

2. **Actionable outputs.** The system produces recommendations, scores, structured diagnoses, roadmaps, or ranked resources – not raw data or open-ended conversation.

3. **Structural intelligence beyond retrieval.** The system reasons over collected data to produce classification, scoring, and gap detection – not merely retrieval or summarisation. Rule-based logic, scoring models, and AI components must work in concert.

4. **Explainability.** Every output is traceable. Users and judges can understand why the system produced a given result, what evidence supports it, and how each score was computed.

5. **Evaluation.** The team defines at least one measurable evaluation metric and runs it on a test set, however small.

1.4 Submission Requirements

1.4.1 First Submission – Concept and Prototype Foundation

The first submission is an early-stage deliverable demonstrating that the team has understood the problem, committed to a technical direction, and begun building.

- **Concept presentation** – problem framing, proposed solution, target users, core AI mechanism, maturity taxonomy draft, and initial scoring dimensions; format is free (slide deck, document, or mockup); maximum 10 pages or slides
- **GitHub repository link** – must contain at minimum: a structured README describing the chosen approach and scope, initial project structure or architecture sketch, and any early code, mockups, or design decisions, even if not yet functional

> **First submission is not a finished prototype.** Judges will evaluate clarity of problem understanding, credibility of the technical direction, and early evidence of execution – not completeness. A well-structured repository with a strong README and a clear scoring methodology draft is more valuable than a working but misdirected prototype.

1.4.2 Final Submission

Every team must submit the following on demo day:

- **Pitch deck** – problem, solution, target users, value proposition, demo walkthrough, limitations, and next steps (maximum 15 slides)
- **Demo video** – a recorded walkthrough of at least one end-to-end scenario demonstrating a core capability of the system (maximum 5 minutes); a live demo may be substituted during the presentation session
- **Architecture diagram** – system components, data flow, AI pipeline, and key cross-module integration points
- **Knowledge base documentation** – sources used, formats, key fields, ingestion pipeline description, and coverage notes
- **Scoring methodology document** – criteria definitions, weights, aggregation logic, and justification for modelling decisions
- **Explainability layer** – a mechanism or interface element showing why the system produced a specific output, including score breakdowns and diagnostic justifications
- **Evaluation report** – metric(s) used, test protocol, and results (even on a small test set)
- **GitHub / GitLab repository** – source code with a clear README and setup instructions

1.5 Non-Functional Expectations

- **Responsiveness.** User-facing queries should return results within a few seconds for a realistic dataset size.
- **Reliability.** The system handles missing, dirty, or incomplete data without crashing – document how edge cases are handled.
- **Privacy & Security.** Any sensitive data (personal, financial, project-specific) must be masked or anonymised. State clearly what data is used and how it is protected.
- **Scalability mindset.** Explain how the system would scale to more data, more users, or a broader geographic scope.

1.6 What Is Not Allowed

- Standalone chatbot or assistant projects – conversational features may exist as a secondary layer connected to the diagnostic and scoring engines, not as the core product. The system must do something structurally non-trivial beyond answering questions.
- Solutions where the conversational assistant *is* the core product. The diagnostic engine, explainable scoring module, and RAG-grounded recommendation layer are mandatory. A chat interface alone – even one backed by a strong LLM – does not meet the requirements.
- Solutions without a clear, direct connection to entrepreneurship in the Tunisian context.
- Reproducing an existing commercial product without a meaningful original contribution specific to this problem space.

> The goal is not to prescribe what you build – it is to ensure you build something real. The use case defines the problem space and the acceptance criteria. Design, architecture, product decisions, and technology choices are entirely yours.

2 Use Case – Intelligent Entrepreneurial Orientation Engine

An AI System for Project Maturity Diagnosis, Explainable Multi-Dimensional Scoring, and Personalised Roadmap Generation

2.1 Problem

Entrepreneurs at the early and growth stages of their journey face a compounding set of orientation failures. They lack visibility over their actual project maturity level, the critical steps they need to take, and the support devices that apply to their situation. The information that exists is fragmented, rarely personalised, and almost never contextualised to a specific project profile. A structural problem compounds this: entrepreneurs consistently overestimate or mischaracterise the maturity of their projects. A founder who believes they are ready for financing may have no validated business model. A project described as scalable may be entirely dependent on manual accompaniment. Without a system capable of detecting these gaps, orientation tools simply reflect and reinforce the entrepreneur's existing – potentially inaccurate – self-perception. Existing tools – static guides, generic chatbots, and disconnected administrative portals – do not address this. They answer questions but do not diagnose. They provide information but do not produce a structured, evidence-based assessment of where a project actually stands and what it actually needs.

2.2 Vision

Build a unified AI platform that places the entrepreneur's project profile at the centre and activates three intelligent capabilities around it: an adaptive diagnostic engine that classifies the project and surfaces gaps; an explainable scoring system that evaluates the project across multiple dimensions with transparent, weighted criteria; and a RAG-grounded recommendation layer that orients the entrepreneur toward real resources, programs, and actions drawn from a curated knowledge base. The integration is the differentiator. A diagnostic finding that identifies a missing market validation step should surface relevant support programs that address it. A scalability gap should connect to specific actions in the roadmap. The three modules must interact – not merely coexist.

> **Architecture note:** The three features below are mandatory. Teams are free to design the integration layer – shared project profile store, unified scoring API, cross-module recommendation logic – in any way that best serves the user experience. What will be evaluated is whether the modules genuinely interact, or merely coexist as independent panels.

2.3 Feature 1 – Adaptive Diagnostic Engine

2.3.1 Problem Scope

Entrepreneurs are rarely able to accurately position themselves within their development cycle. They either overestimate their readiness – claiming to be at a financing stage without a validated model – or underestimate the structuring work already done. No existing tool in the Tunisian entrepreneurial ecosystem performs a dynamic, evidence-based diagnostic that combines structured data collection, adaptive questioning, maturity classification, and explicit gap detection between self-perception and project reality. The result is misaligned orientation: entrepreneurs are sent to programs they do not qualify for, offered resources that do not match their actual stage, and left without a clear picture of what is genuinely blocking their progress.

2.3.2 Data & Domain Hints

- **Maturity stage taxonomy:** A classification framework covering at minimum six stages, listed in chronological order: (1) **Ideation**, (2) **Market Validation**, (3) **Structuration**, (4) **Fundraising** (formerly pre-financing), (5) **Launch Planning**, and (6) **Growth** – with defined criteria per stage
- **Entrepreneur profile dimensions:** Sector, legal form, team composition, revenue status, validation evidence, business model clarity, geographic location, prior accompaniment history
- **Gap detection cases:** Synthetic or documented cases illustrating perception-reality divergence (e.g. self-assessed "financing-ready" vs. actual diagnostic output of "structuration")
- **Blocker taxonomy:** A structured catalogue of common entrepreneurial blockers mapped to maturity stages and domains (financial, legal, market, technical, organisational)

> **Design note:** The adaptive intake is not a static form. Questions must evolve in response to prior answers – a project flagged as operating in the agri-food sector should trigger sector-specific questions; a project claiming market traction should trigger validation evidence probes. The intelligence of the intake is part of what will be evaluated.

2.3.3 Expected Capabilities

- **Adaptive structured intake** – collect project and entrepreneur profile information through a dynamic questionnaire whose branching logic responds to prior inputs
- **Maturity classification** – assign the project to a stage in a defined taxonomy, with evidence linking classification to collected data points
- **Gap detection** – identify and surface discrepancies between the entrepreneur's declared self-assessment and the system's classification; flag specific dimensions causing the divergence
- **Blocker identification** – detect and rank priority blockers based on collected data; distinguish between structural gaps and missing information
- **Diagnostic synthesis** – produce a structured, readable diagnostic output – not a raw score – that the entrepreneur can act on
- **Contextual project memory** – persist the project profile across sessions so that the diagnosis refines over time as new information is added

2.3.4 Acceptance Criteria

Criterion	What to Demonstrate	Priority
Adaptive intake is real	Branching logic produces meaningfully different question sequences for at least 3 distinct profiles	Must
Classification is traceable	Every maturity stage assignment links to specific collected data points	Must
Gap detection works	At least 3 demonstrated cases where self-assessed stage diverges from system diagnosis	Must
End-to-end demo	Intake to diagnostic output runs without manual intervention	Must
Blocker identification	Priority blockers are ranked and linked to the maturity stage	Should
Handles ambiguity	Incomplete profiles are handled gracefully; uncertainty is surfaced rather than hidden	Should
Persistent project context	Re-entering the system with updated information refines the diagnosis	Should
Evaluation protocol	At least one classification metric reported on a labelled test set	Should

2.3.5 Key Considerations

- **Classification accuracy matters.** Misclassifying a project stage can send an entrepreneur down the wrong path. Be explicit about confidence levels and edge cases.
- **Perception gap is a feature, not a side effect.** The ability to detect divergence between self-assessment and actual diagnosis is a core differentiator – design for it explicitly, not as an afterthought.
- **Language.** The intake should operate in French and/or Arabic to be accessible to the primary user population.
- **Scope discipline.** Define a focused maturity taxonomy with clear, defensible criteria. Six stages with clear boundaries are more valuable than ten ambiguous ones.

2.4 Feature 2 – Explainable Multi-Dimensional Scoring

2.4.1 Problem Scope

Evaluating an entrepreneurial project requires reasoning across multiple incommensurable dimensions simultaneously – market potential, commercial offer strength, innovation intensity, and scalability are not reducible to a single axis. An entrepreneur should be able to understand not just *whether* their project scores well, but *why* – which specific criteria are driving each sub-score, what the gaps are, and what concrete actions would improve a given dimension. Without this granularity, a score is not a decision-support tool; it is a label.

2.4.2 Scoring Dimensions

Teams must implement at least the following **five** composite scores. Each composite score must decompose into explicit sub-scores with defined, weighted criteria.

Score	Sub-dimensions to cover
Market Score	Market share (combining addressable market size and competitive landscape analysis); customer validation evidence; revenue model clarity and viability
Commercial Offer Score	Clarity and differentiation of the value proposition; product/service maturity and readiness; pricing strategy coherence; alignment between offer and identified customer needs
Innovation Score	Local novelty and differentiation; technology intensity; barrier to entry; degree of departure from existing market offerings
Scalability Score	Replicability without linear cost increase; manual accompaniment dependency; deployment cost structure; addressable market size and geographic potential
Green Score	Sustainability and environmental impact evaluation; project's ecological viability and carbon footprint mitigation; alignment with Sustainable Development Goals (SDGs); resource efficiency and circular economy practices

2.4.3 Expected Capabilities

- **Weighted criteria model** – each sub-score is computed from a defined set of criteria with explicit weights; weights are documented and justifiable
- **Score decomposition** – the composite score and each sub-score are surfaced to the user with per-criterion contributions
- **Natural-language justification** – each score is accompanied by a plain-language explanation of what drove the result
- **Anomaly and inconsistency detection** – the system flags profiles with contradictory or unsubstantiated signals (e.g. a project claiming high market traction with no validation evidence, or high scalability with heavy manual dependency)
- **Improvement guidance** – for each score, the system identifies the highest-leverage gap and suggests a concrete action that would improve it
- **Score evolution** – as the project profile is updated, scores recalculate and changes are tracked

2.4.4 Acceptance Criteria

Criterion	What to Demonstrate	Priority
Five composite scores implemented	Market, Commercial Offer, Innovation, Scalability, and Green scores all computed and displayed	Must
Sub-scores are explicit	Each composite score decomposes into at least 3 sub-dimensions with visible contributions	Must
Criteria weights are documented	Weighting methodology is described and defensible	Must
Natural-language justification	Every score accompanied by a plain-language explanation	Must
Anomaly detection works	At least 2 demonstrated cases of contradictory or unsubstantiated signals flagged	Should
Improvement guidance is specific	Highest-leverage gap per score identified with a concrete suggested action	Should
Score evolution tracked	Profile update triggers score recalculation with change highlighted	Should
Evaluation protocol	Scoring consistency or inter-rater agreement measured on a test set	Should

2.4.5 Key Considerations

- **Explainability is not optional.** A score without a traceable justification is not an explainable score – it is an opaque label. Every criterion contribution must be surfaced.
- **Scoring is modelling work.** Defining the criteria, weights, and aggregation logic is a significant part of this feature. Teams must document their modelling decisions – this will be evaluated as part of Technical Depth.
- **Composite scores are not averages.** The aggregation logic for each composite score should reflect the domain – a weak score on a fundamental dimension should not be masked by strong scores elsewhere.

2.5 Feature 3 – RAG-Grounded Roadmap & Resource Orientation

2.5.1 Problem Scope

Once a project has been diagnosed and scored, the entrepreneur needs to know *what to do next* and *where to go for support*. Generic advice is not sufficient. The recommendations must be grounded in real, available resources – support programs, financing devices, administrative procedures, and ecosystem actors that actually exist in Tunisia – and they must be personalised to the specific diagnostic output and scoring profile of this entrepreneur. The core failure of existing orientation tools is that their recommendations are either generic (applicable to any entrepreneur) or untraced (plausible-sounding but not grounded in a real source). A recommendation that cannot be linked to a real program, document, or resource is not a recommendation – it is content generation.

2.5.2 Knowledge Base Construction

Teams must build and document a real knowledge base. This is core work, not optional enrichment. The knowledge base should cover at minimum:

- **Support and accompaniment programs** – APII, BFPME, BTS, Startup Act mechanisms, ANPE resources, incubators and accelerators active in Tunisia
- **Financing programs** – AFD programs, EU funding mechanisms, bank financing products, UNDP acceleration programs, international development fund mechanisms applicable in Tunisia
- **Administrative and regulatory resources** – relevant procedures, forms, and institutional contacts for business creation and operation in Tunisia
- **Business guides and standards** – certification pathways, quality frameworks, and reporting standards applicable in the MENA region
- **Ecosystem actors** – mentors, networks, and professional associations relevant to entrepreneurship in Tunisia

> **Data note:** No single clean dataset for this knowledge base exists. Part of the challenge is ingesting, structuring, and indexing heterogeneous documents – PDFs, web pages, structured lists – into a corpus that a retrieval pipeline can exploit. How this pipeline is designed and documented is part of what will be evaluated.

2.5.3 Expected Capabilities

- **RAG pipeline** – retrieve relevant resources from the knowledge base in response to the diagnostic profile and scoring output; every retrieved item must be traceable to its source
- **Personalised roadmap generation** – produce an ordered, prioritised action plan grounded in the diagnostic output; actions at different time horizons (immediate, short-term, medium-term)
- **Resource matching** – surface support programs, financing devices, and administrative steps relevant to the entrepreneur's current stage and identified gaps
- **Cross-module coherence** – a diagnostic gap should trigger retrieval of relevant support resources; a low sub-score should surface knowledge base items that address that dimension
- **Dashboard restitution** – a visual interface presenting maturity level, composite scores with sub-score breakdowns, priority blockers, recommended next actions, and the roadmap
- **'Mon Parcours" tracking view** – a persistent view where the entrepreneur sees their current stage, past recommendations, actions taken, and next steps; updates as the project profile evolves
- **Connected conversational assistant** – a secondary conversational layer that responds to questions using the diagnostic results, scores, roadmap, and knowledge base as its grounding context; it does not operate independently of these structured outputs

2.5.4 Acceptance Criteria

Criterion	What to Demonstrate	Priority
Knowledge base is real	At least 30 documented, real resources structured and indexed	Must
Retrieval is traceable	Every recommended resource cites at least one source in the knowledge base	Must
Roadmap is personalised	Different diagnostic outputs produce meaningfully different roadmaps	Must
Cross-module coherence	A diagnostic gap or low sub-score demonstrably triggers relevant knowledge base retrieval	Must
Dashboard is functional	Maturity level, scores, blockers, roadmap all visible in a single interface	Must
'Mon Parcours" view exists	Persistent tracking view updates with project profile evolution	Should
Conversational assistant is grounded	Assistant responses reference diagnostic results, scores, or knowledge base items – not generic LLM output	Should
Evaluation protocol	Retrieval relevance or roadmap coherence metric reported on a test set	Should
Knowledge base is updatable	New resources can be added without rebuilding the retrieval pipeline	Could

2.5.5 Key Considerations

- **Knowledge base construction is core work.** Building a structured, diverse, and documented knowledge base of real Tunisian resources is a significant deliverable. Plan for it in your timeline – it cannot be improvised on demo day.
- **Recommendations must be grounded.** A recommendation that cannot be traced to a specific item in the knowledge base or a specific output from the diagnostic engine is not acceptable. Hallucinated program names or invented administrative steps constitute a critical failure.
- **The assistant is a layer, not the product.** The conversational assistant must be demonstrably connected to the structured outputs of the other two features. An assistant that answers questions from general knowledge, bypassing the diagnostic and knowledge base, does not meet the requirement.
- **Roadmap is not a list.** A personalised roadmap has an order, a rationale, and time horizons. A flat list of generic actions is not a roadmap.

3 Evaluation & Judging Framework

3.1 Judging Criteria

All projects will be evaluated by a panel of judges including PNUD / GEWEET representatives, technical AI experts, and entrepreneurship practitioners. Scores are assigned across five dimensions:

Dimension	What Judges Look For	Weight
Real-world Impact	Direct relevance to entrepreneurship in Tunisia, clarity of the value proposition, plausibility of real adoption	25%
Technical Depth	Quality of the AI pipeline, intelligence of the diagnostic and scoring engines, non-trivial problem solving beyond keyword search or chatbot interaction	25%
Prototype Quality	End-to-end demo works, handles realistic inputs, produces meaningful outputs, interface usable by a non-technical entrepreneur	20%
Explainability & Scoring Rigour	Scores are traceable to weighted criteria; diagnostic classifications are linked to evidence; users can understand why any output was produced; system communicates uncertainty	15%
Evaluation & Rigour	Team defined and ran a real evaluation protocol with documented metrics, a test set, and reported results	15%

3.2 Bonus Points

- **Cross-module integration depth** – the three mandatory features interact meaningfully around the shared project profile: diagnostic gaps trigger knowledge base retrieval; low sub-scores surface targeted roadmap actions; the assistant references structured outputs
- **Perception-reality gap detection is non-trivial** – the system correctly flags at least 3 cases where entrepreneur self-assessment diverges from the diagnostic output, with a clear explanation of the divergence
- **Real user validation** – at least one real potential user (entrepreneur, support program officer, or investor) tested the prototype and provided documented feedback
- **Arabic language support** – the system handles Arabic-language inputs or knowledge base documents meaningfully – not just translation
- **Original dataset contribution** – the team built, curated, and documented a dataset that did not exist before: a structured Tunisian entrepreneurial support program catalogue, a maturity stage taxonomy with labelled examples, or a sector-specific knowledge base
- **Post-hackathon roadmap** – a credible, specific plan for continuing development toward real adoption with the PNUD / GEWEET ecosystem